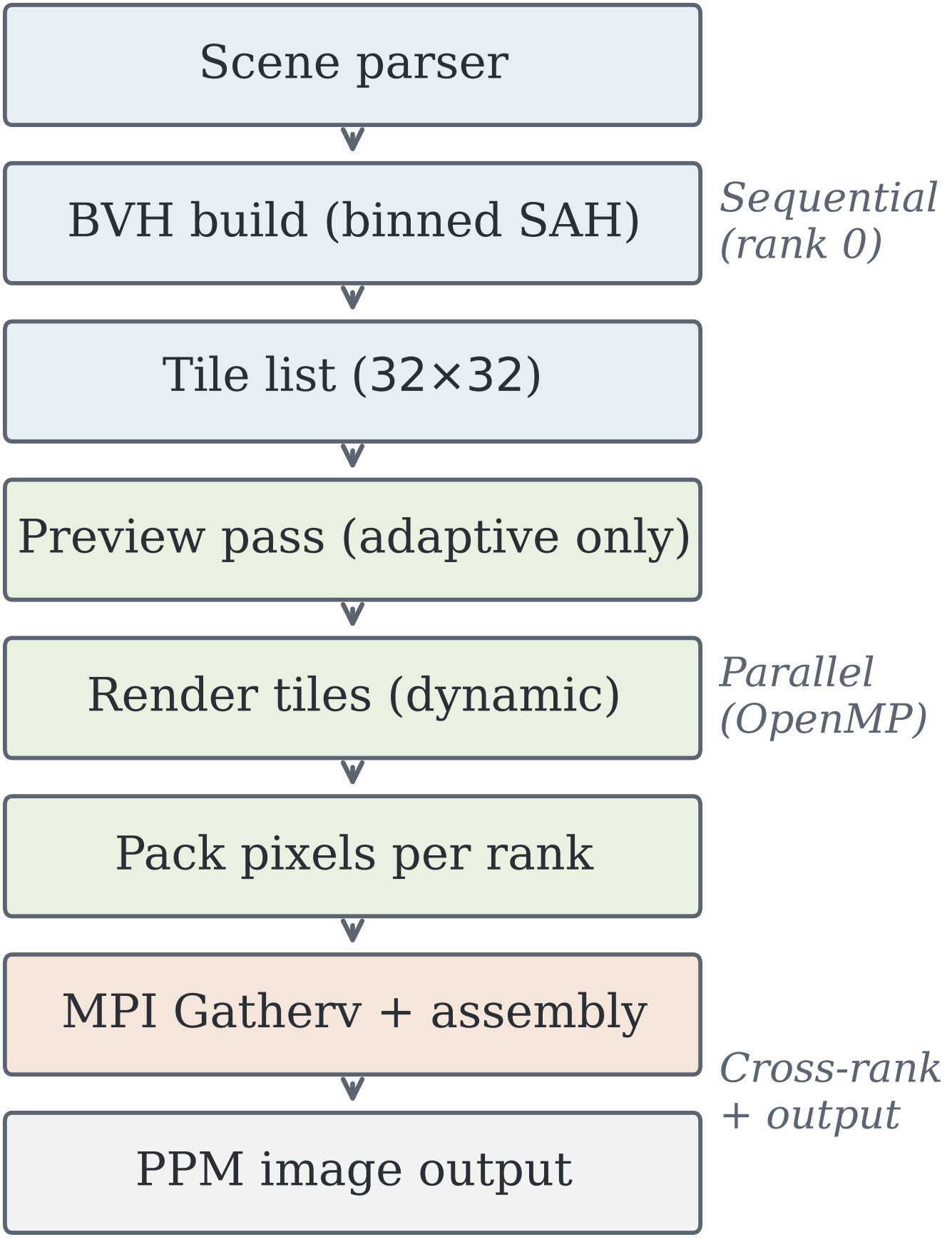
Scene parser
BVH build (binned SAH)
Sequential (rank 0)
Tile list (32×32)
Preview pass (adaptive only)
Render tiles (dynamic)
Parallel (OpenMP)
Pack pixels per rank
MPI Gatherv + assembly
Cross-rank + output
PPM image output